

- 20** A fixed mass of ideal gas at constant pressure occupies a volume V . The gas undergoes a rise in temperature so that the root mean square velocity of its molecules is doubled.

What is the new volume?

A V

B $\sqrt{2}V$

C $2V$

D $4V$

20	D	Internal energy = kinetic energy = $\frac{3}{2}pV = \frac{1}{2}m \langle c^2 \rangle$ When rms velocity doubles, internal energy increases by 4 times. Since pressure is constant, V increases by 4 times. (internal energy for ideal gas is $\frac{3}{2} pV$)
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- 21** In the diagram, the volume of bulb X is twice that of bulb Y . The system is filled with an ideal